

DHARANIKOTA NAGA DEEPIKA

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Summary

AI/ML undergraduate skilled in Python, React.js, Node.js, NLP, Generative AI, and AI agent creation. Experienced in building LLM-powered applications and end-to-end ML systems using TensorFlow, Scikit-learn, FastAPI, and AWS.

Skills

Programming: C, C++, Python, JavaScript, TypeScript, Node.js

Frontend: React.js, HTML, CSS

AI/ML: Machine Learning, Deep Learning, Reinforcement Learning, NLP, Generative AI, Computer Vision, TensorFlow, Scikit-learn

Cloud & Tools: AWS EC2, S3, CloudFront

Global Professional Certifications

AWS Certified Cloud Practitioner (CLF-C02)

Jun 2026

Experience

AI/ML Virtual Intern

April 2025 – June 2025

AICTE-EduSkills (AWS Academy)

- Built a cloud-based **Stroke Risk Prediction** app using **Scikit-learn**, **FastAPI**, **Docker**, and **AWS**.

Projects

SkillPath AI – Adaptive Learning Coach | *Reinforcement Learning, Python, NumPy, Streamlit*

- **AI Agent:** Built from scratch Deep Q-Network (NumPy) for adaptive difficulty/topic recommendation based on real-time learner performance.
- **MDP & Reward Design:** Designed the RL environment — state space, action space, and reward function — with live online training via experience replay.
- **Model Evaluation:** Built an evaluation harness benchmarking the DQN against baseline policies using paired statistical testing.

AI-Powered Question Paper Generator | *Python, Scikit-learn, NLP, Pandas, NumPy*

- **NLP Pipeline:** Classified questions by topic and difficulty from PDFs using TF-IDF and Scikit-learn; achieved **90% accuracy**.
- **Auto Generation:** Produced balanced question papers by topic and Bloom's taxonomy level, reducing manual effort.

Context-Aware Notification Filter | *Python, FastAPI, Node.js, Scikit-learn, RL*

- **Smart Filtering:** Applied ML and RL-based ranking to prioritize notifications and reduce spam dynamically.
- **Adaptive Prediction:** Applied machine learning and reinforcement learning techniques for dynamic ranking and filtering.
- **Backend:** Built a FastAPI + Node.js REST API consumed by web and Android clients.

AI Smart Parking & Navigation System | *Python, YOLOv8, OpenStreetMap*

- **Detection & Navigation:** YOLOv8 for real-time parking slot detection; OpenStreetMap APIs for GPS-mapped route guidance.
- **Spatial Mapping:** Converted image detections into GPS coordinates for map-based tracking and navigation.
- **Geospatial Integration:** Used OpenStreetMap APIs for parking visualization and route guidance.

Awards & Certifications

- **Top 2% Nationally (2025):** NPTEL Elite+Gold – *The Joy of Computing using Python*.
- **Winner – SheHack:** AI-powered Question Bank Generation and Smart Question Paper Authoring System.

Education

Gayatri Vidya Parishad College of Engineering

B.Tech in Computer Science and Engineering (AI/ML), Expected 2027

Visakhapatnam, AP

CGPA: 9.3/10